63.

**Solution :**

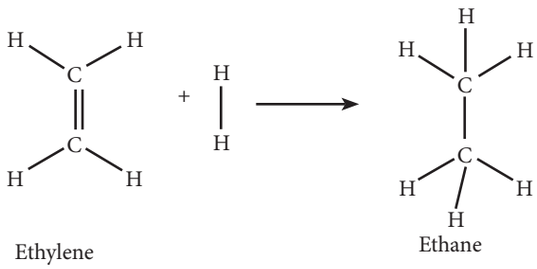
**Given :**

$E_{C-H} = 414 \text{ kJ mol}^{-1}$

$E_{C-C} = 347 \text{ kJ mol}^{-1}$

$E_{C=C} = 618 \text{ kJ mol}^{-1}$

$E_{H-H} = 435 \text{ kJ mol}^{-1}$



Ethylene Ethane

$\Delta H_r = \Sigma (\text{Bond energy})_r - \Sigma (\text{Bond energy})_p$

$\Delta H_r = (E_{C=C} + 4E_{C-H} + E_{H-H}) - (E_{C-C} + 6E_{C-H})$

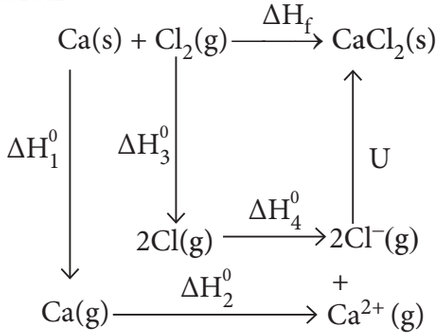
$\Delta H_r = (618 + (4 \times 414) + 435) - (347 + (6 \times 414))$

$\Delta H_r = 2709 - 2831$

$\Delta H_r = -122 \text{ kJ mol}^{-1}$

64.

**Solution :**




$\Delta H_f = \Delta H_1 + \Delta H_2 + \Delta H_3 + 2\Delta H_4 + u$

$-795 = 121 + 2422 + 242.8 + (2 \times -355) + u$

$-795 = 2785.8 - 710 + u$

$-795 = 2075.8 + u$

$u = -795 - 2075.8$

$u = -2870.8 \text{ kJ mol}^{-1}$

65.

**Solution :**

**Given :**

$\Delta H_f (Fe_2O_3) = -741 \text{ kJ mol}^{-1}$

$\Delta H_f (CO) = -137 \text{ kJ mol}^{-1}$

$\Delta H_f (CO_2) = -394.5 \text{ kJ mol}^{-1}$

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

$\Delta Hr = ?$

$\Delta Hr = \Sigma(\Delta H_f)_{\text{products}} - \Sigma (\Delta H_f)_{\text{reactants}}$

$\Delta Hr = [2 \Delta H_f (Fe) + 3\Delta H_f (CO_2)] - [\Delta H_f (Fe_2O_3) + 3\Delta H_f (CO)]$

$\Delta Hr = [0 + 3 (-394.5)] - [-741 + 3 (-137)]$

$\Delta Hr = [-1183.5] - [-1152]$

$\Delta Hr = -1183.5 + 1152$

$\Delta Hr = -31.5 \text{ kJ mol}^{-1}$



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